Lab 5 Setup Guide: Private S3 Bucket behind CloudFront

This guide deploys the Lab 5 resources with one CloudFormation template, then walks through checking that the bucket is private and the site is served only through CloudFront.

What the template creates

Resource	Purpose
S3 bucket	Private. All four Block Public Access settings are on and nothing turns them off.
Origin Access Control	CloudFront's signed identity when it reads from the bucket.
CloudFront distribution	Serves index.html at the root over HTTPS, using the bucket as its origin.
Bucket policy	Lets the CloudFront service read objects (s3:GetObject), only for this one distribution.

The bucket is named <stack name>-<BucketSuffix>-<account ID>, so it is globally unique. The default suffix is mission-ui.

Before you start

- Permissions to create S3, CloudFront and CloudFormation resources.
- The AWS CLI (CloudShell works), and the lab's built site files, including an index.html.
- CloudFront is a global service, so the stack can be created in any region.

Part 1: Deploy the stack

1. Save the template (the full text is in the appendix) as cloudfront-private-bucket-lab.yaml.
2. Deploy it. Replace leap-lab5 with your stack name: (leap-lab5-teamX-firstName-lastName)

```
aws cloudformation deploy \
  --template-file cloudfront-private-bucket-lab.yaml \
  --stack-name <stack-name>
```

3. Read the outputs:

```
aws cloudformation describe-stacks --stack-name <stack-name> \
  --query "Stacks[0].Outputs" --output table
```

The distribution can take several minutes to finish creating. Wait for the stack to report CREATE_COMPLETE.

Part 2: Upload the site

4. Copy your index.html into the bucket, using the BucketName output:

```
aws s3 sync ./build s3://<BucketName>/
```

5. Confirm index.html is at the top level of the bucket, not inside a folder.

Part 3: Verify the Deployment

A Navigate to the index.html file in your bucket and click ‘open’. Try both the object URI and the S3 URI as well.

Why this works

- Block Public Access stays on, so no one can open the bucket to the world by mistake.
- The bucket policy names the CloudFront service as the only principal, not "*". S3 therefore does not treat it as a public policy.
- The condition on AWS:SourceArn limits access to this single distribution. Another distribution cannot read the bucket.
- The origin is the bucket's regional address, not the S3 website address. The website endpoint cannot be private.

Cleanup

A bucket that still contains files cannot be deleted by CloudFormation, so empty it first:

1. Delete the index.html file from the bucket.

2. Initiate a delete of the resources for the stack:

```
aws cloudformation delete-stack --stack-name <stack-name>
```

The distribution is disabled and deleted as part of the stack; this can take several minutes.

In the AWS Console, go to CloudFormation and your associated stack name should show its in the process of being deleted.